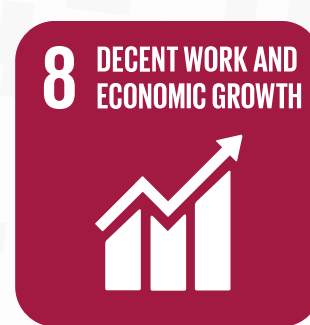




Farm Market

EMPOWERING YOUNG FARMERS THROUGH DIGITAL TRADE



Abstract

FARMARKET is a localized digital marketplace system designed to connect young farmers in Pampanga directly with consumers, reducing reliance on middlemen and promoting fair and transparent agricultural trade. The system was developed using an Agile-based System Development Life Cycle (SDLC) and integrates key features such as secure user access, product listing, order management, and real-time communication. A mixed-methods approach was employed to evaluate the system using the Technology Acceptance Model (TAM),

focusing on perceived usefulness, perceived ease of use, and behavioral intention. Results indicate very high user acceptance among young farmers, highlighting improvements in market access, selling efficiency, and income potential. Despite challenges related to internet connectivity and digital literacy, FARMARKET demonstrates strong feasibility as a user-centered and sustainable digital solution that supports local agricultural digitalization in Pampanga.

Keywords

FARMARKET, digital marketplace, young farmers, Technology Acceptance Model, agricultural digitalization

Objectives

Platform Performance and Reliability

Assess system stability, security, and core marketplace functionality.

User Acceptance and Usability

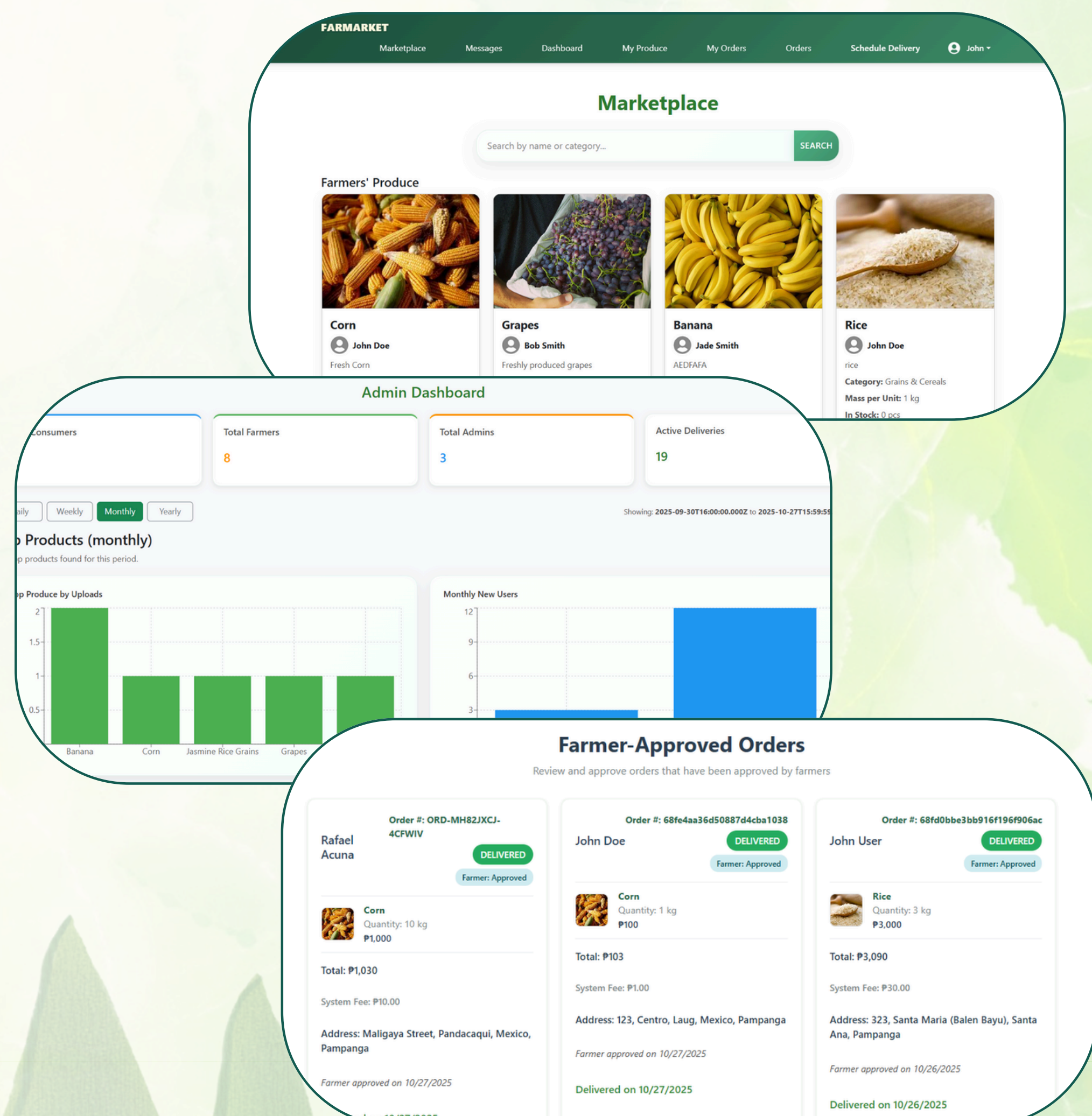
Evaluate perceived usefulness, ease of use, and adoption intention using TAM.

Economic and Community Impact

Examine improvements in market access, income potential, and middlemen reduction.

System Effectiveness Evaluation

Validate system usability and quality through surveys and user feedback.



Methodology

- Requirements Analysis – Identification of young farmers' challenges and system requirements
- System Design – Design of user-centered marketplace, dashboards, and secure communication features
- Development – Iterative implementation using Agile sprints
- Testing – User acceptance testing with young farmers and IT experts
- Evaluation – System assessment using Technology Acceptance Model (PU, PEU, BI) and descriptive statistics
- Refinement – Continuous system improvement based on user feedback

Conclusion

The study confirms the need for a localized digital marketplace that supports fair and transparent agricultural trade. FARMARKET demonstrates high usability, strong user acceptance, and adoption potential by enabling direct farmer-to-consumer transactions and reducing middlemen dependency. While limitations in internet access and digital literacy remain, the system is validated as a practical digital solution that can enhance income opportunities and promote agricultural digitalization in Pampanga.

Results

- Improved market access, transparency, and fair pricing for young farmers
- Direct farmer-to-consumer transactions reduced middlemen dependency
- Very high perceived usefulness, ease of use, and ease of learning
- The system addressed key agricultural marketing and selling needs
- The platform supports efficient selling, better coordination, and local economic growth

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Lawrence Manuel M. Baluyut
Researcher

Rafael Francis T. Acuña
Researcher

John Carlo M. Dizon
Researcher

Fernand T. Layug
Researcher Adviser

Grace D. Tiqui
Researcher Adviser

